

Global Climate Change Adaptation Research (1990–2025): Trends, Methodologies, and Emerging Research Gaps

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Abstract – As environmental dangers and socioeconomic vulnerabilities increase worldwide; climate change adaptation has emerged as a critical field of study. In order to identify important trends, methodological developments, prevailing theme areas, and current research shortages, this study offers a thorough evaluation of worldwide climate change adaptation research carried out between 1990 and 2025. About 150 excellent papers from major academic databases and worldwide institutional repositories were examined using bibliometric analysis in conjunction with a systematic literature review that adhered to PRISMA principles. The findings show that adaptation research has undergone a substantial change, moving from early conceptual frameworks and risk assessments in the 1990s to sophisticated, data-driven, and multidisciplinary approaches in the 2020s. In order to improve climate resilience planning, recent research have increasingly used machine learning, geospatial technology, predictive modelling, and integrated policy frameworks. Coastal ecosystem preservation, urban resilience, water resource management, agriculture and food security, and health adaption are some of the main study topics. The worldwide research landscape is still unequal despite significant advancements, with a large concentration of papers coming from wealthy nations and under-representation from extremely vulnerable regions including South Asia, Africa, and tiny island states. Furthermore, there are still significant gaps in the integration of indigenous knowledge systems, social fairness, real-time climate data, and the efficient conversion of scientific discoveries into practical policy. This study highlights the significance of egalitarian, AI-supported, and transdisciplinary methods and offers a conceptual framework that connects research progress with real-world implementation procedures in order to address these issues. In order to advance inclusive and successful climate change adaptation plans globally, it is imperative that global collaboration and data accessibility be strengthened.

Keywords – *Climate Change, Climate Adaptation, Sustainable Development, Climate Resilience, Bibliometric Analysis, Research Trends, Environmental Sustainability, Global Warming.*

1. Introduction:

A significant worldwide phenomenon that has an impact on social, economic, and environmental systems is climate change. Global surface temperatures have risen by around 1.1°C since the pre-industrial era, according to the Intergovernmental Panel on Climate Change (IPCC), increasing the frequency and severity of extreme weather events. Ecosystems and human civilizations are negatively impacted by these effects, especially in poor nations with low ability for adaptation [6].

Since its rise to popularity in the early 1990s, climate adaptation research has developed into an interdisciplinary subject that incorporates data analytics, environmental science, socioeconomic studies, and governance. The topic has seen tremendous expansion in scholarly publications, international cooperation, and the creation of adaptation plans between 1990 and 2025 [8].

Early research mostly concentrated on conceptual resilience frameworks and vulnerability assessment. To address region-specific adaptation issues, recent research has increasingly used sophisticated approaches including machine learning, predictive modelling, and remote sensing. Examining trends and approaches in adaptation research has become more important due to the increasing frequency of climate-related disasters, such as floods, droughts, cyclones, and wildfires.

In order to identify important themes, methodological advancements, and future research paths that promote efficient policy formulation and resilience planning, this study examines global climate adaptation research conducted during the previous 35 years.

2. SIGNIFICANCE OF THE STUDY:

This work advances knowledge of how global adaptation research has evolved to address interrelated environmental and socioeconomic issues. A thorough understanding of the evolution and maturation of adaptation frameworks throughout time may be obtained by examining patterns from 1990 to 2025 [8][9].

Key Significance Points:

1. **Bridging Knowledge Gaps:** Provides a comprehensive understanding of adaptation science by connecting disparate studies from different fields and geographical areas.
2. **Policy Relevance:** Offers vital information to promote adaptive governance for national and international policymakers, such as the IPCC, UNEP, and UNFCCC.

3. **Scientific Advancement:** Emphasizes how sophisticated computational and data-driven techniques have replaced qualitative approaches, increasing the rigor of research.
4. **Regional Prioritization:** identifies understudied areas with significant adaptation requirements, such as South Asia, the Pacific Islands, and sub-Saharan Africa [9].
5. **Integration of Local Knowledge:** Encourages the integration of contemporary scientific methods with indigenous and traditional methods for inclusive adaptation.
6. **Future Preparedness:** Describes a study agenda with a focus on action-oriented, data-driven, and transdisciplinary adaptation studies.

3. REVIEW OF LITERATURE:

Linzhi Ou (2025): Ou examines planning for climate adaptation in Chinese metropolitan areas, emphasizing adaptive governance, low-carbon development, and spatial adaptation. The study finds substantial policy advancements using bibliometric data (1990–2023), CiteSpace [1], and policy text mining, but it also exposes a lack of integration of climate resilience in urban planning, especially in AI-based modelling and community involvement.

Achenafi Teklay (2025): Teklay investigates how smallholder farmers in Ethiopia's Megech Watershed perceive and respond to climatic unpredictability [2]. The study uses 30 years of climate data, focus groups, and 385 household surveys to find a high level of knowledge of rising temperatures and decreasing rainfall, which are frequently connected to industrial expansion and deforestation. Major obstacles, such as high input costs, poor extension services, and restricted credit, are identified via logistic regression and descriptive analysis. The findings highlight the necessity of regional adaptation strategies to increase smallholder resilience in midland areas.

Andrew G. Rundle (2025): Rundle's systematic review synthesizes 31 studies across 24 countries to examine the impact of climate change on educational outcomes. The findings show reduced learning time, school attendance, and cognitive performance, with stronger effects in marginalized populations. Using PRISMA-ScR and CHANGE frameworks, five key education domains affected by climate risks are identified. The study concludes that climate change amplifies educational inequities, particularly in low- and middle-income countries with limited data [3].

Francesca Roseo (2025): Using PRISMA-based techniques and modelling tools, Roseo examines 88 papers (2011–2025) to assess the effects of climate change on mountain ecosystem services. Mountains are important hotspots for biodiversity and water supplies, and the study reveals that over half of the consequences are unfavorable, especially when it comes to supplying, regulating, and cultural functions. Strong regional bias toward Europe and the Tibetan Plateau is also revealed by the study, and little research has been done on supporting services in understudied mountain regions [4].

Haimanote K. Bayabil (2025): By examining smallholder farmers' responses to climate change at three locations in Ethiopia's Megech Watershed, Bayabil expands on Teklay's research. According to the study, which used regression models and structured surveys, 87% of farmers implemented strategies including better seed types and modified planting dates. Ownership of livestock, financial availability, and educational attainment are found to be important determinants of adaption success. The results emphasize the necessity of location-specific analysis to evaluate obstacles and the efficacy of adaptation in midland areas [5].

Perry G. B. de Louw (2025): De Louw uses KNMI climate scenarios and hydrological modelling to assess the sustainability of rainwater lenses in the Southwest Delta of the Netherlands. Despite more winter rainfall, the data indicate that freshwater lens loss will increase owing to drier summers. The study suggests low-water crops and drainage optimization as adaptation strategies, and it presents five metrics to evaluate lens stability. It also highlights the need of including unsaturated hydrology and soil salinity into plans for coastal adaptation [6].

America-van den Heuvel et al. (2025): Using numerical sensitivity analysis in the Dutch delta, this companion study to de Louw's research verifies the shallowing and growing susceptibility of freshwater lenses. By examining seepage flux, soil characteristics, and crop types, the study highlights drainage optimization as a key resilience strategy. Additionally, it suggests using salinization and dynamic crop development models to enhance future resilience forecasts [7].

Debernardi et al. (2023): Using NLP and topic modelling, the study examines regional research specialization in 193,471 climate change articles from 1990 to 2020. The findings indicate a decentralization of climate research, with affluent countries emphasizing policy and theoretical work and poor countries increasingly conducting applied studies. Highly susceptible areas are still under-

represented despite the identification of five significant geographic clusters, underscoring the need for fair research funding and international involvement [8].

Albugami et al. (2024): Albugami et al. use Bibliometric R and VOSviewer to examine 1,411 Scopus articles (2004–2023) about climate change and sustainable lifestyles in South Asia. Bangladesh and Pakistan are the next biggest contributors, after India. Vulnerability, adaptability, and sustainability are important issues. The study maps collaboration networks and emerging trends but highlights the need for causal analyses and multidisciplinary approaches to inform evidence-based policy [9].

Wei et al. (2024): Wei et al. evaluate the evolution of carbon sinks by analyzing 1.4 billion hectares of agriculture worldwide between 1982 and 2018. The study uses spatial overlay techniques and structural equation modelling to determine that around 30% of the increase in carbon sinks may be explained by increased CO₂, mostly through GPP-driven processes. While pointing out the need to look into long-term soil salinization, the effects of harsh climate, and non-grain carbon routes, the authors conclude that farmland makes a considerable contribution to carbon neutrality [10].

Gautam & Shany (2024): The study examines 40 years of climate data from 30 districts in Tamil Nadu, India, using the Hargreaves-Samani method and CMIP6 projections. Results indicate increasing post-monsoon aridity and flash flood risk, particularly in semi-urban areas. The study links semi-urbanization to higher carbon emissions and recommends urgent policy interventions for water management, emphasizing the integration of indigenous knowledge into adaptation frameworks.

MacLean et al. (2024): MacLean et al. examine 82 research (1998–2023) to evaluate the effects of climate change on alcohol and drug users. According to the study, extreme weather conditions including heat, floods, and droughts harm mental health, interfere with the provision of services, and increase substance abuse. It creates a framework that connects climatic events to addiction patterns, identifies gaps in low-income nations, and suggests climate-adaptive addiction strategies using narrative synthesis and scoping techniques.

Sponagel et al. (2024): The study uses life cycle assessment models and farm-level geodata to examine agrivoltaics (AV) systems in Baden-Württemberg, Germany. According to the results, installing AV on 1–5% of agricultural land might still be viable while reducing GHG emissions by

up to 5.9 Mt CO₂-eq. AV is a cost-effective decarbonization technique, according to the PALUD-LCA hybrid model, but further study is needed to determine its scalability and effects on biodiversity.

Hickinbotham et al. (2025): Hickinbotham uses topic modelling and co-word analysis to examine 4,681 studies (1989–2019) on phenology connected to climate change. The study exposes regional and taxonomic biases, with the majority of studies concentrated on Northern Hemisphere agriculture, birds, and insects. Despite its widespread usage, remote sensing and ecological modelling are still not well linked. In order to enhance predictive ecological research, the report advocates for transdisciplinary techniques, especially in the Global South.

Ajibade et al. (2023): The bibliometric study looks at machine learning applications in climate science by analyzing 2,706 Scopus papers from 2008 to 2022. Water resources, meteorology, agriculture, and catastrophe prediction are recognized as the four primary hotspots. Leading contributions (USA, China) are highlighted by co-authorship and keyword network analyses, and the study highlights the need for more financing and multidisciplinary cooperation in poorer nations to advance ML-driven climate research.

Raihan (2023): The most susceptible industries include agriculture, health, biodiversity, and tourism, according to Raihan's analysis, which summarizes global climate concerns. Using a comparative review paradigm, the paper suggests integrated government-led climate policy and connects climatic variability to the development of illness and food shortages. However, successful implementation in developing countries is hampered by a lack of empirical evidence and policy assessment.

Purica & Ali (2024): The study creates maps of flood and drought risk using temperature and precipitation data from Romania from 1961 to the present. To allocate insurance and mitigation resources in accordance with the EU 2030 climate plan, a data-driven model is put forth. While pointing out inadequacies in global standardization and multi-variable integration across areas, the paper offers a scalable framework by combining temperature variance analysis with spatial risk modelling.

Hinz et al. (2024): Hinz evaluates the effects of climate change on fish communities by analyzing 45 years' worth of ecological data from Palma Bay, Spain. Nine species are at high danger because

of rising sea temperatures, according to a climate risk score that incorporates species' thermal tolerances and habitat preferences. To differentiate anthropogenic from climatic effects, the research suggests long-term monitoring and bioenergetics modelling.

Trummer et al. (2023): Trummer's qualitative study examines the health effects of migration brought on by climate change using webinars and expert interviews conducted in Sudan and Zimbabwe. Results indicate that gender-based vulnerabilities and issues with mental and reproductive health are made worse by migration. The study uses a community-of-interest method to identify gaps in gender-sensitive adaption techniques, mental health therapies, and climate-health-migration policies in African contexts.

De Louw et al. (2025): These studies cross-check freshwater lens deterioration research conducted along the Dutch coast. It offers reliable estimates of salinity and hydrological stability under various climatic scenarios using a data-driven sensitivity model. Nevertheless, there are still limitations in the creation of AI-based forecasting systems for the sustainability of coastal freshwater.

Overall Synthesis (1990–2025): Data analytics, adaption science, and sustainability frameworks have all made significant strides in climate change research during the past thirty years. Regression modelling, scoping reviews, bibliometric mapping, and geographical analysis are common techniques. There are still gaps in multidisciplinary integration, developing-country representation, and AI/ML applications for climate resilience. The trend highlights the need for equitable, data-driven, system-level adaptation solutions globally by shifting from descriptive to predictive techniques.

4. PROBLEM AREAS IN EXISTING SYSTEM

Global adaptation research is limited by a number of factors, despite notable advancements:

1. **Fragmented Approaches:** Rather than being system-integrated, adaptation studies are still mostly discipline-specific (e.g., hydrology vs. social science) [6].
2. **Regional Imbalance:** Less than 15% of research comes from South Asia and Africa, two extremely susceptible regions, compared to over 65% from North America and Europe [8][9].

3. **Limited Data Availability:** Regional modelling and repeatability are hampered by inconsistent and closed-access resources.
4. **Weak Science-Policy Linkage:** There is a gap between knowledge and execution since findings are rarely translated into workable policy [16].
5. **Lack of Technological Integration:** Adaptation studies underutilize cloud-based modelling, big data, and artificial intelligence.
6. **Under-representation of Local Communities:** Despite the advantages of resilience, indigenous and community-based knowledge is still underutilized.
7. **Institutional and Financial Barriers:** Inadequate institutional capability and a lack of climate funding make adaption strategies less scalable.

5. OBJECTIVES OF THE STUDY

This study's main goal is to compile research on global climate adaptation from 1990 to 2025, highlighting significant developments, methodological breakthroughs, and enduring knowledge gaps that have shaped the subject.

The following primary and secondary goals serve as the study's compass:

A. To Analyze the Chronological Trends in Global Climate Change Adaptation Research (1990–2025)

- Use bibliometric measures, such as publication numbers, citations, and cooperation networks, to assess the progress of adaptation research.
- Determine times of fast growth, especially after international accords (e.g., Kyoto Protocol 1997, Paris Agreement 2015).
- Monitor changes from risk reduction and vulnerability assessment in the 1990s to transformational adaptation and resilience building after 2010.
- Evaluate how global research orientations are impacted by policy developments, climate disasters, and geopolitical events.

B. To Classify and Evaluate Major Research Methodologies Used in Adaptation Studies

- Chart the transition from qualitative case studies to quantitative, model-based, and data-driven approaches, such as machine learning, GIS, and climate modelling.
- Examine how cross-disciplinary integration has affected results by comparing disciplinary methods (environmental science, economics, social policy, and engineering).
- Examine how community-based and participatory approaches might improve the implementation of local adaptation.
- Since 2015, highlight cutting-edge approaches including network analysis, bibliometric mapping, and multi-criteria decision-making frameworks.

C. To Identify Key Themes, Regions, and Sectors Covered in Adaptation Research

- Group research into the following main themes: urban resilience, water management, health, coastal adaptation, and agriculture and food security.
- Draw a map showing geographic concentrations, highlighting the dominance of North America and Europe and the under-representation of South Asia, Africa, and Latin America.
- Evaluate sectorial variety, ranging from ecosystem-based adaptation to technical advancements like resilient renewable energy and climate-smart agribusiness.
- To emphasize integrated adaptation planning, look at cross-sectoral connections, such as water-energy-food (WEF) nexus research.

D. To Highlight Existing Research Gaps and Future Opportunities for Interdisciplinary Studies

- Draw attention to theme gaps, such as the lack of focus on gender, migration, and health adaption.
- Point out methodological flaws, such as a lack of long-term research and a restricted application of big data or real-time climate modelling.
- High-risk areas, such as the Pacific Islands, Sahel, and Himalayan ecosystems, are under-represented, exposing geographic gaps.
- Take note of the shortcomings in governance and policy, particularly the poor conversion of research into local implementation.

- To increase prediction ability and policy relevance, future research should promote interdisciplinary integration of environmental, social, and data sciences.

E. To Propose Recommendations for Strengthening Global Adaptation Strategies and Policies

- Create frameworks to convert research into policy so that planning and decision-making for adaptation are informed by scientific evidence.
- Encourage the creation of open-access global adaptation databases that incorporate socioeconomic, environmental, and climate-risk data.
- Encourage capacity-building programs to enable institutions and researchers in developing nations to produce knowledge in a balanced manner.
- Promote climate financing systems that give evidence-based adaptation initiatives top priority.
- Put in place decision-support tools with AI and GIS capabilities to track the efficacy of adaption in real time.
- To improve cultural, ecological, and social sustainability, incorporate indigenous and local knowledge into formal adaptation frameworks.

F. To Develop a Conceptual Framework Linking Research Evolution with Practical Implementation

- Create a conceptual model that connects policy implementation, methodological development, and research trends.
- To facilitate ongoing progress, show feedback loops between research findings and practical adaption outcomes.
- Offer academics, governmental organizations, and international organizations engaged in climate-resilience planning a strategic guidance.

G. Expected Outcome of Objectives

By fulfilling these goals, the research will:

- Describe the development of global adaptation research in a methodical, fact-based manner. Provide a multidisciplinary taxonomy of approaches and subject areas.
- Determine which understudied areas and fields need funding.

- Make practical suggestions to improve the link between science and policy.
- Contribute to the global knowledge base by proposing a replicable framework for evaluating adaptation research in future decades.

6. RESEARCH METHODOLOGY AND DATA COLLECTION

The study uses a methodical, analytical, and descriptive approach that combines qualitative content synthesis, systematic literature review (SLR), and bibliometric analysis. Finding, evaluating, and interpreting patterns and the development of global climate adaptation research between 1990 and 2025 is its goal.

A. Research Design

This non-experimental, exploratory research describes patterns, correlations, and trends without changing any of the variables. The design consists of:

- Descriptive analysis, which summarizes theme trends and publishing patterns.
- Analytical synthesis: determining research clusters and methodological progress.
- Comparative analysis, which evaluates theme and geographical differences.

To guarantee repeatability and transparency, the process adheres to PRISMA criteria.

B. Data Sources: Secondary data from reliable international academic and institutional sources is used in the study.

Table I: Data Sources and their purpose for Global Climate Adaptation Research

Source Type	Database / Repository	Purpose
Academic Databases	Scopus, Web of Science, Google Scholar	Retrieve scholarly articles, journal papers, and reviews.
Climate Reports	IPCC, UNEP, UNFCCC	Extract adaptation policy frameworks and trends.

Institutional Repositories	World Bank, UNDP, NASA Earth Data	Access environmental and socio-economic datasets.
Journals	Nature Climate Change, Global Environmental Change, Climatic Change	Collect peer-reviewed research publications.

C. Data Collection Process

The study used an organized procedure to collect and filter data:

Keyword Search: used phrases like "adaptive governance," "climate policy," "sustainability," "vulnerability," "resilience," and "climate change adaptation."

1. Time Frame: concentrated on works of literature released between 1990 and 2025.

2. Inclusion Criteria:

- Conference papers and peer-reviewed journal publications.
- Publications written in English.
- Research that focuses on methods, tactics, or evaluations for climate change adaptation.

3. Exclusion Criteria:

- Studies focused solely on mitigation.
- Policy or non-scientific commentary that lacks empirical support.
- Low-impact or duplicate papers.

7. DATA SCREENING AND VALIDATION:

- Duplicate entries were manually eliminated.
- Verified accuracy by cross-referencing data from other databases.
- Final dataset: around 150 core publications were examined for thematic and bibliometric trends.

A. Data Analysis Tools and Techniques

To guarantee thorough and transparent analysis, the study used both quantitative and qualitative methodologies.

Table II: Tools and Software's used for Quantitative and Qualitative Analysis

Tool/ Software	Purpose of Use
VOSviewer	Visualization of bibliometric networks—authors, institutions, countries, and keywords.
CiteSpace	Mapping citation clusters, co-authorship, and temporal trends.
Excel / SPSS	Statistical analysis and data categorization.
NVivo	Thematic and qualitative content analysis.

B. Data Parameters Considered

- Publication Year: To chart trends over time.
- Authors and Institutions: To pinpoint important contributors.
- Geographic Focus: The distribution of adaption studies by region.
- The methodology employed might be mixed, qualitative, or quantitative.
- Sectorial Themes: urban, agricultural, water, health, or adaptation based on policy.
- Keywords: To find new study themes, use frequency analysis.

C. Research Ethics

Citations and references were kept in accordance with APA requirements, and all data came from publicly accessible sources to ensure ethical compliance.

Methodology

The operational processes and analytical techniques are described in this section. To guarantee thorough coverage and scientific rigor, the investigation was carried out in six methodical steps.

Phase 1: Literature Identification

- Boolean operators (AND, OR, NOT) were used in keyword-based searches to find literature. An example query might be "adaptation to climate change" AND "vulnerability" AND "resilience."
- Over 1,000 records were initially retrieved from Scopus and Web of Science.

Phase 2: Screening and Selection

- There were about 350 papers left after duplicates and irrelevant research were eliminated.
- Each manuscript was examined for methodological clarity, abstract substance, and title relevancy.
- For full-text analysis, a final batch of 150 excellent publications was chosen.

Phase 3: Data Coding and Classification

- Every study was categorized and assigned a code by:
- Thematic area: policy, health, urban systems, agriculture, water, and coastal ecosystems.
- Research approach: mixed, qualitative, quantitative, or model-based.
- Regional emphasis: developed vs developing nations.
- Publication source: Reputable publications and organizations.
- Results and gaps: Manually extracted and coded in order to analyze patterns.

This coding made it easier to compare and visualize adaption patterns in an organized manner.

Phase 4: Data Analysis

1. Bibliometric Evaluation

Recognized growth trends, prolific writers, and partnerships.

VOSviewer is used to visualize co-authorship and co-citation networks.

2. Analysis of Themes

NVivo analyses abstracts and keywords to find new ideas.

Resilience, governance, vulnerability, and adaptation planning were common themes.

3. Content Analysis

Qualitative assessment of 150 studies provided insights into methodologies, policy integration, and practical applications.

Phase 5: Model Development

A framework was proposed linking research evolution → knowledge generation → policy adaptation → implementation outcomes.

- The model highlights feedback loops between research and governance to strengthen adaptive capacity.

Phase 6: Synthesis and Reporting

- Combined all analytical findings to find trends in adaption studies throughout time.
- Emphasized policy, methodology, and data deficiencies.
- Made suggestions to improve global adaptation governance and trans disciplinary cooperation.

Methodological Justification

- Global research performance is measured by bibliometric analysis.
- Deeper patterns and meanings are captured via thematic and qualitative analysis.

A thorough grasp of the evolution of adaptation research may be obtained by combining the two.

8. DATA COLLECTION

In order to guarantee accuracy and dependability, the study uses secondary data, with a focus on reliable, authentic, and peer-reviewed sources.

A. Data Collection Approach

In order to methodically find, screen, and evaluate pertinent studies, the study used an SLR in accordance with PRISMA principles. Important actions included:

1. Finding the sources of the data.
2. Selection and screening according to inclusion/exclusion standards.
3. Data extraction, both qualitative (textual) and quantitative (bibliometric).
4. Information gathered is validated and categorized.

B. Data Sources

The study did not gather any primary data; instead, it depends on secondary data. Sources consist of:

- Academic databases: Google Scholar, Web of Science, and Scopus.
- Reports from the World Bank, UNEP, IPCC, and UNDP.
- Government portals: country-level policy papers and National Adaptation Plans (NAPs).

- Climate databases: NOAA, NASA Earth Data, and World Climate provide contextual information.

C. Data Extraction Process

- Bibliographic information was extracted, including author, year, title, keywords, nation, and journal.
- Annual publication counts and citations are quantitative measures that have been documented.
- Combined thematic variables: locale, adaption type, and research emphasis.
- Excel and VOSviewer were used to store and analyze data for cross-comparison.

D. Data Organization

Three levels were created using the collected data:

1. Macro-level: Worldwide patterns and trends (such as an increase in publications).
2. Meso-level: Sectoral representation and regional dispersion.
3. Micro-level: Case studies and particular approaches.

This hierarchical structure made it possible to monitor the advancement of research at various sizes.

E. Data Validation

- Data from several databases was cross-checked.
- Duplicate and low-impact papers were eliminated.
- Verified the legitimacy of the journal using the Impact Factor Index (IFI) and Scimago Journal Ranking (SJR)

F. Challenges Faced During Data Collection

- Restricted publicly available statistics from underdeveloped nations.
- Variations in terminology between research (such as "adaptation," "vulnerability," and "resilience"). Geographic inclusiveness decreased when non-English media were excluded.
- Pay walled databases result in time and access limitations.

G. Strengths of Data Collection Process

- A thorough trend analysis is made possible by the 35-year longitudinal coverage.
- Reliability is improved via multi-database source.

- Human prejudice is lessened by structured coding.
- Integration of bibliometric and qualitative content provides a holistic dataset.

H. Summary

The framework ensured methodical organization, authenticity, and academic integrity throughout the study by offering a strong basis for bibliometric mapping, trend analysis, and theme synthesis.

9. RESEARCH PLAN

Through a planned, phased strategy, this research, "Global Climate Change Adaptation Research (1990–2025): Trends, Methodologies, and Emerging Research Gaps," takes a methodical, analytical, and descriptive approach. The plan provides a strategic blueprint guiding activities from conceptualization to synthesis, ensuring methodological consistency and academic rigor. Stages progress logically from problem identification to data analysis and interpretation, forming a coherent pathway linking objectives, methodology, and outcomes.

Phase 1: Conceptualization and Topic Formulation

The first step of the study is to determine the central issue, which is research trends and approaches related to global climate change adaptation. The researcher looks at new scientific fields, environmental issues, and the global setting. The study's applicability to climate science and sustainability research is supported by a problem diagnosis. To make sure they are in line with academic viability, time restrictions, and existing literature and data, the title and objectives are modified.

Phase 2: Literature Review and Theoretical Foundation

A thorough analysis of research from 1990 to 2025 was carried out, encompassing groundbreaking studies, journal articles, and reports from organizations like the IPCC, UNEP, and UNFCCC. This review highlights conceptual gaps and methodological constraints while identifying important theoretical models, frameworks, and empirical results on adaptation and resilience. It also supports the selected approach and provides information for the study framework.

Phase 3: Data Collection and Compilation

Bibliometric and qualitative data were gathered from academic databases and institutional repositories, such as Scopus, Web of Science, SpringerLink, and Google Scholar, after the literature review. There were no fieldwork or original data collected. A verified dataset for further research was created by downloading the metadata, abstracts, and keywords of pertinent studies, filtering them according to inclusion criteria (peer-reviewed, adaptation-specific, published 1990–2025).

Phase 4: Data Analysis and Interpretation

The gathered data is analyzed both quantitatively and qualitatively in this step. Collaboration networks, publishing trends, keyword frequencies, and citation patterns may all be found using bibliometric tools (VOSviewer, CiteSpace). Key concepts, approaches, and areas of interest are interpreted through qualitative theme analysis. Combining the two methods offers a thorough understanding of conceptual advancements and numerical patterns. The development of adaptation research is shown by visual maps and thematic clusters, which emphasize important subjects including vulnerability, resilience, policy integration, and adaptation funding in addition to regional and sectorial research needs.

Phase 5: Model Development and Synthesis

A conceptual model connecting the development of adaptation research to real-world implementation outcomes was created based on analytical findings. The model, which is based on empirical patterns, shows how organized scientific research influences resilience theories, adaption tactics, and policy. It highlights the circular nature of adaptation research, where research informs policy, which in turn directs additional exploration and innovation, by integrating ideas from environmental science, economics, and policy studies.

Phase 6: Report Writing and Documentation

The last stage gathers research results into an organized report that includes an introduction, a review of the literature, a methodology, an analysis, a discussion, and a conclusion. Academic

rigor is ensured via editing, citation checking, and APA-compliant reference. The study summarizes 35 years of adaptation research, emphasizing both ongoing gaps and successes. Policymakers, researchers, and institutions are guided toward integrated, data-driven adaptation methods via recommendations. Key outcomes include:

- Comprehensive synthesis of 35 years of adaptation research.
- Identification of thematic, methodological, and regional trends.
- A conceptual model bridging research and policy.
- Contribution to the discourse on sustainable adaptation and climate resilience.
- A reference resource for researchers, students, and policymakers in global environmental governance.

10. CONCLUSION

According to the study's findings, research on global climate change adaptation has grown from conceptual frameworks in the 1990s to sophisticated, data-driven, multidisciplinary investigations in the 2020s. Despite advancements, research is still unequal, with industrialized countries dominating and social, technical, and indigenous viewpoints not being integrated. Future research should use AI-based analytical tools, improve cooperation, and close the gap between policy implementation and scientific discoveries. Sustainable development in the face of climate change requires inclusive, data-driven, and locally appropriate adaption measures.

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